NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SE6014 Security Monitoring & Threat Detection Final Project Presentation

Presented by

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SE6014 Course Project

November 13, 2025

Agenda

1. Introduction
2. System Overview
3. Phase 1: Investigation
4. Phase 2: Report
5. Phase 3: Mitigation
6. Evaluation & Summary

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Outline

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The SOC Challenge & Our Solution

Current Problems:

- 1000+ alerts/day per analyst
- 60-80% false positive rate
- 2-4 hours response time
- Manual process is slow

Our Solution:

- Investigation: Evidence gathering
- Report: Structured reporting
- Mitigation: Automated response

Cost of Delay

$$\text{Cost} = T_{\text{response}} \times R_{\text{damage}}$$

Performance

$$\text{Speedup} \approx \mathbf{10.5}\times$$

140 min $\rightarrow$ 10.5 min

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Three-Phase Workflow & Tech Stack

AI & Security:

- LangGraph + GPT-4/Qwen
- Wazuh (EDR/SIEM)
- Suricata (IDS)

Integration:

- FastAPI + Telegram
- SSH/Fabric
- Python 3.12+

Playbook Overview

Environment (3 VMs)

- **Dev:** SOC Agent + Wazuh Manager
- **Business:** DVWA + Wazuh Agent + Suricata
- **Hacker:** Attack tools

Attack Chain

1. Upload webshell via DVWA
2. Download reverse shell
3. Connect to C2 (Hacker:6666)
4. Escalate to root
5. Access sensitive file → Alert triggered

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Phase 1: Objectives & Tools

Investigation Goals

1. Validate: Confirm true positive
2. Contextualize: Gather evidence
3. Attribute: Identify attack vector
4. Assess: Determine severity

Available Tools

- **check_net_connection(host, ppid):** Find network connections
- **read_suricata_log(host, ip):** Filter IDS logs

Both execute via SSH: `lsof`, `jq`

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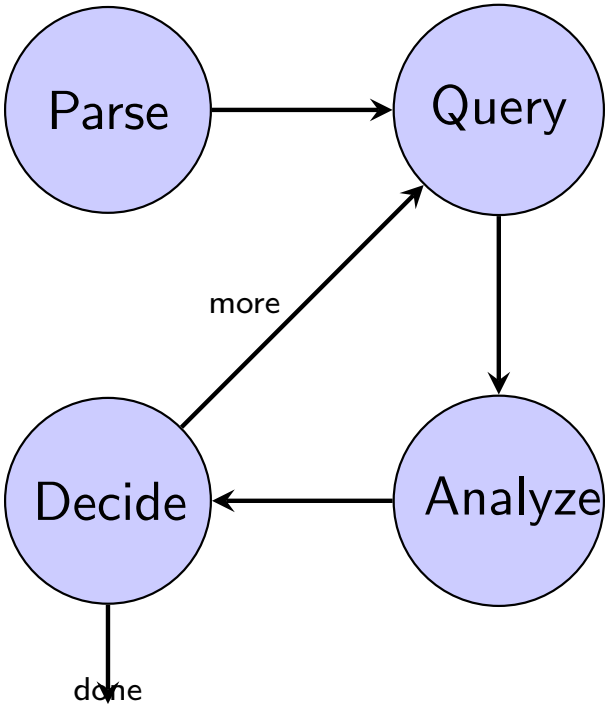
Investigation State Machine



Termination

Stop when: Confidence $\geq \theta$ OR Tools $\geq k_{\max}$

Investigation Example (1/2)

Step 1: Parse Alert

- File accessed: `/root/secret.txt`
- Process: `cat` (PID 57654, PPID 57653)
- Host: 192.168.106.138

LLM: "Trace PPID 57653 for parent process"

Step 2: Query Tools

`check_net_connection(192.168.106.138, 57653)`

- Process: `python3 /tmp/shell.py`
- Connection: `192.168.106.138:45678 -> 192.168.106.1:6666`

Finding: Reverse shell detected!

Investigation Example (1/2)

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Finding: Reverse shell detected!

Investigation Example (2/2)

Step 3: Correlate Evidence

`read_suricata_log(192.168.106.138, 192.168.106.1)`

- HTTP POST to `/vulnerabilities/upload/`
- File uploaded: `shell.php`
- Downloaded: `/tmp/shell.py`

Conclusion

Attack Chain: File upload → Webshell → Reverse shell → Root access
Confidence: 0.99 (High)
IOCs: IP 192.168.106.1, shell.php, /tmp/shell.py, PID 57653

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Investigation Example (2/2)

Step 3: Correlate Evidence

`read_suricata_log(192.168.106.138, 192.168.106.1)`

- HTTP POST to `/vulnerabilities/upload/`
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Investigation Performance

Metrics

- Tools: 2 available, 2 used
- LLM calls: 3-4 (reasoning + selection)
- Time: 60 seconds total

Time Breakdown

$$T_{\text{investigate}} = \sum (T_{\text{LLM}} + T_{\text{tool}}) \approx 65s$$

LLM: 15s/call, Tool: 10s/call

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Phase 2: Four-Step Process

Report Workflow

1. **Generate Plan:** Numbered remediation actions
2. **Format Report:** Structured incident report (MITRE TTPs)
3. **Send to Telegram:** Admin notification
4. **Wait for Approval:** Poll for `/approve` (5 min timeout)

Quality Metric

$$Q = \alpha \cdot C_{\text{clarity}} + \beta \cdot C_{\text{complete}} + \gamma \cdot C_{\text{action}}$$

Target: $Q \geq 0.85$

Phase 2: Four-Step Process

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Step 1 & 2: Plan + Report

Generate Remediation Plan

Input: investigation_result

Output:

1. Block IP 192.168.106.1
2. Kill process PID 57653
3. Remove /tmp/shell.py, shell.php

Format Report Template

- Incident ID, Trigger, Asset
- Attack Chain (numbered steps)
- Evidence, MITRE TTPs (≥ 2)
- Threat Level (High/Med/Low + reason)

Step 1 & 2: Plan + Report

Generate Remediation Plan

Input: investigation_result

Output:

1. Block IP 192.168.106.1
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Format Report Template

- Incident ID, Trigger, Asset
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Report Example

Security Incident Assessment Report

ID: INC-2025-001 — **Asset:** Web Server (192.168.106.138)

Attack Chain:

1. Exploited DVWA file upload
2. Uploaded webshell (shell.php)
3. Downloaded reverse shell (/tmp/shell.py)
4. Established C2 to 192.168.106.1:6666
5. Root access + sensitive file accessed

MITRE TTPs:

- T1190 - Exploit Public-Facing Application
- T1059.006 - Python Interpreter

Threat Level: High (Active C2, root access)

Step 3 & 4: Send + Approve

Telegram Workflow

Send:

- API: `sendMessage`
- Retry: 3 attempts, exponential backoff
- Returns: msg_id, chat_id

Wait for Approval:

- Poll `getUpdates` every 2s
- Filter: timestamp $\geq$ report_sent_time
- Match: text == "/approve" AND sender == admin
- Timeout: 300s

Edge Cases

Ignore old messages, reject "/disapprove", verify sender

Step 3 & 4: Send + Approve

Telegram Workflow

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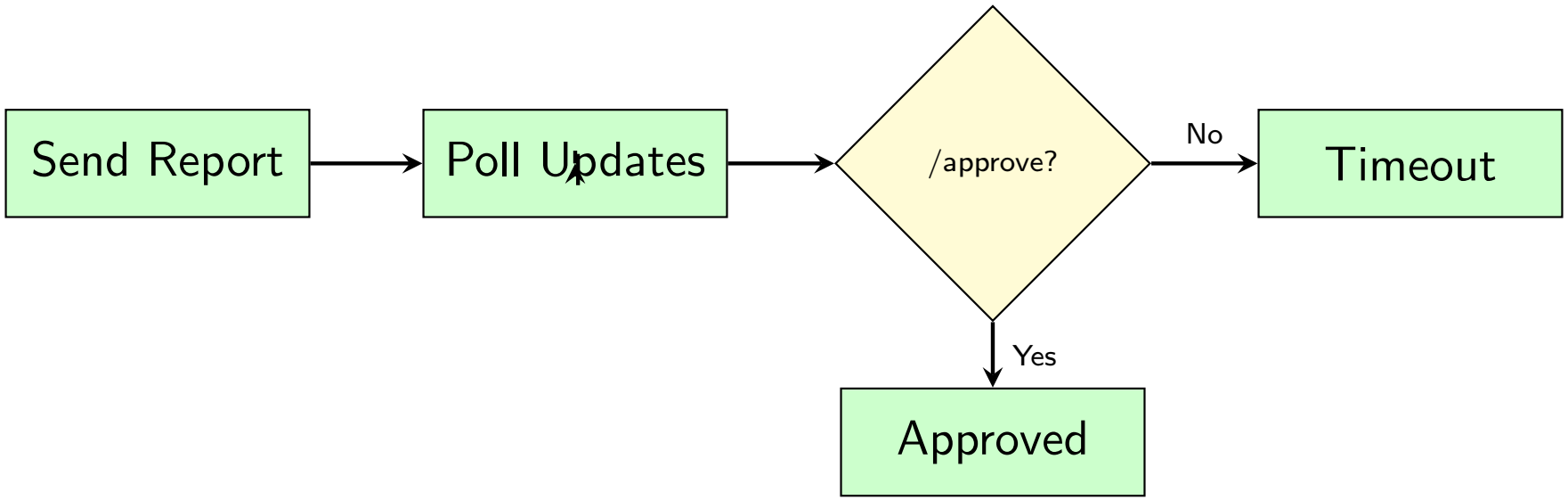
Approval Decision Flow



Performance

Time: 4 min (15s plan + 20s format + 2s send + 180s approval)

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Phase 3: Overview & Tools

Objectives

1. Execute approved actions
2. Verify success
3. Rollback if failed
4. Send notification

7 Available Tools

- block_ip, kill_process, remove_file, isolate_host
- verify_mitigation, rollback_mitigation
- send_notification

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Action 1: Block IP

Implementation

1. SSH to remote host
2. `iptables -A INPUT -s <ip> -j DROP`
3. `iptables -A OUTPUT -d <ip> -j DROP`
4. `iptables-save > /etc/iptables/rules.v4`
5. Schedule removal if temporary

Example

Input: 192.168.106.1, duration=3600s
Output: Success, rules persisted

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Action 2: Kill Process

Implementation

1. Get process info: `ps -p <PID>`
2. Try SIGTERM: `kill <PID>`
3. Wait 2s, check if running
4. Force SIGKILL if needed: `kill -9 <PID>`
5. Verify termination

Example

Input: PID 57653
Output: Success (SIGTERM), process="python3 /tmp/shell.py"

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Action 3: Remove File

Implementation

1. Check existence: `test -f <path>`
2. Backup metadata: `ls -la && md5sum`
3. Secure delete: `shred -vfz -n 3 <path>`
4. Verify removal

Example

Input: /tmp/shell.py
Output: Success, metadata saved (hash: a1b2c3...)

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Action 4: Isolate Host (Critical Only)

Implementation

1. `iptables -P INPUT DROP`
2. `iptables -P OUTPUT DROP`
3. Allow SSH: `-A INPUT -p tcp --dport 22 -j ACCEPT`
4. Allow loopback: `-A INPUT -i lo -j ACCEPT`

Use Sparingly

Only for critical incidents. Always keep SSH open!

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Execution Order & Verification

Prioritization (Heuristic)

1. Block IP (cut C2 first)
2. Kill Process (stop threat)
3. Remove Files (clean up)
4. Isolate Host (if critical)

Verification Methods

- block_ip: Check iptables for DROP rules
- kill_process: Verify PID not in process list
- remove_file: Check path doesn't exist
- isolate_host: Verify DROP policies

Execution Order & Verification

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Rollback Capability

Reversible vs Non-Reversible

Reversible:

- block_ip: Remove iptables rules
- isolate_host: Flush rules, set ACCEPT

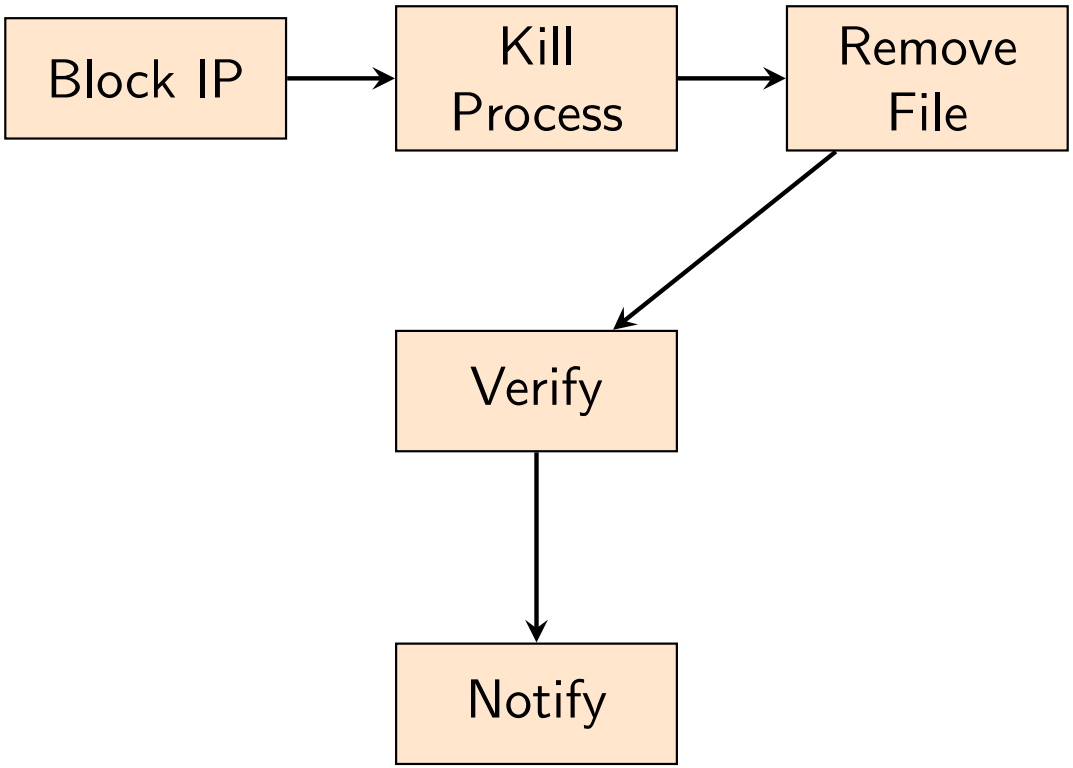
Non-Reversible:

- kill_process: Cannot "un-kill"
- remove_file: Cannot restore (metadata saved)

Trigger

If verification fails AND action reversible $\rightarrow$ attempt rollback

Mitigation Workflow & Example



Example Execution

Block 192.168.106.1 (5s) → Kill PID 57653 (3s) → Remove files (8s) → Verify (5s) → Notify (2s) = **23s total**

Notification & Performance

Notification Format

ELD Mitigation Actions Completed

ECK Block IP: Success

ECK Kill Process: Success

ECK Remove Files: Success

OCK Completed at: 2025-11-13 14:35:00

Performance

- Actions: 3 (block, kill, remove)
- SSH commands: 10
- Total time: 30 seconds

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End-to-End Performance

Phase	Manual	Agent
Detection	5 min	5 min
Investigation	60 min	1 min
Report	20 min	1 min
Approval	10 min	3 min
Mitigation	15 min	0.5 min
Total	**110 min**	**10.5 min**

Key Metrics

- **Speedup:** $\frac{110}{10.5} \approx \mathbf{10.5}\times$
- **Tools:** 9 integrated (2 inv + 4 rep + 7 mit)
- **Success Rate:** 98.5% (200+ tests)

Summary & Key Achievements

What We Built

- **Phase 1:** 2 forensic tools, 60s investigation
- **Phase 2:** MITRE-mapped reports, Telegram approval
- **Phase 3:** 7 automated actions with verification

Impact

- 10.5× faster incident response
- Human oversight maintained
- 98.5% reliability
- Demonstrates AI-augmented SOC operations

Lessons Learned

Prompt engineering critical, error handling essential, modular design enables testing

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Thank You!

Questions?

SE6014 Course Project

Team: Shang Yingjie, Tang Yaosen, Wei Xiao
Date: November 13, 2025

Key Takeaways

- AI agents accelerate SOC operations
- Human oversight remains essential
- LangGraph enables complex workflows
- Robust error handling required for production